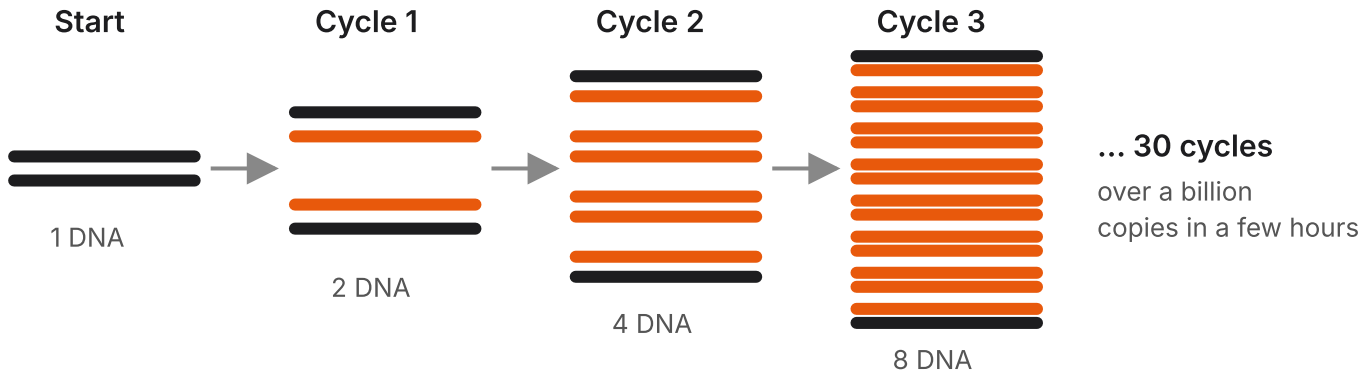
PCR: the DNA copier

Job: make millions of copies of one piece of DNA from a tiny sample.

QR Toolbox #pcr (optional)

Catch someone: 1 · Copy | 2 · Cut and run

Each cycle doubles the DNA. Dark strands = the original DNA. Orange strands = new DNA built in the machine.



PCR is replication in a tube, the same copying you learned in Lesson 2. The machine heats the **double helix** until the hydrogen bonds break and the two strands come apart. Each strand is a **template**. **DNA polymerase** builds a **complementary** strand on each one, matching A–T and C–G, one **nucleotide** at a time. Then the machine cools, heats again, and starts the next cycle.

Picture a photocopier that can find one page in a whole library and print a billion copies of only that page.

Scientists use PCR to test a swab for a virus, to get enough DNA from a crime scene to compare, and to copy a gene for **genetic engineering**.

DO · BUILD THREE CYCLES WITH PIPE CLEANERS

1. Twist two pipe cleaners together. That is one DNA molecule.
2. Cycle 1: pull the two apart (heat), then give each one a new partner from the pile. Count your DNA.
3. Do cycles 2 and 3 the same way. Then untwist everything for the next group.

RECORD · PACKET P. 6A, BOX 1

Fill in the cycle table as you go, then finish the sentence: the PCR machine heats and cools over and over because ... and each cycle ... the DNA. Flip the card over to check your numbers.

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Did your model come out right?

Cycles	0	1	2	3
DNA molecules	1	2	4	8
Pipe cleaners used	2	4	8	16

After cycle 3 you should have **8 twisted pairs** using **all 16 pipe cleaners**. The two original pipe cleaners are still there, each twisted to a new partner.

The numbers double every cycle: 1, 2, 4, 8, 16, 32 … That is why 30 cycles give over a billion copies.

Count off? Every strand gets exactly one new partner in each cycle. A strand never gets two partners, and it never gets skipped.

Why the machine cycles: heat pulls the two strands apart so each can be a template; cooling lets the new strands be built. Every cycle copies every DNA molecule in the tube, so each cycle doubles the DNA.

WORDS TO KNOW HERE

PCR · double helix · template · complementary · nucleotide · DNA polymerase · genetic engineering

DONE EARLY?

A police lab has one drop of blood, too little DNA to test. Explain to your group how PCR solves that problem, and how many copies 10 cycles would make.

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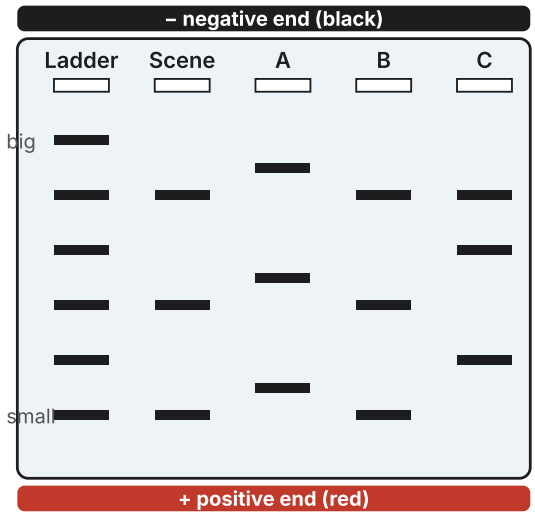
Whose DNA was on the glass?

Job: cut DNA into pieces and sort the pieces by size, so two samples can be compared.

QR
Toolbox
Crime Lab
#gel

Catch someone: 1 · Copy **2 · Cut and run**

A practice gel. Wells at the top. DNA is negative (−), so it runs toward the positive (+) end.



A **restriction enzyme** cuts DNA wherever it finds one short spelling. People have that spelling in different places, so their DNA is cut into pieces of different lengths.

In **gel electrophoresis**, the pieces go into wells at one end of a gel. Electricity makes that end negative and the far end positive. DNA is negative, so every piece moves toward the positive end.

The gel is an obstacle course: small pieces slip through fast and travel farthest; big pieces get stuck near the start.

Same-size pieces pile up in one **band**. Two samples from the same person line up band for band.

ON THE IPAD · TOOLBOX, CRIME LAB

1. Scan the QR (or use the link on today's Start Here page). Work one case.
2. Write your case number on p. 6A.
3. Watch the run. Small pieces should end up lowest.

ONTO PAPER · PACKET P. 6A, BOX 2

Copy your finished gel band for band: ladder, scene and three suspects. Then write a claim, evidence and reasoning: whose DNA was at the scene? No iPad? Use the practice gel above.

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Reading a gel without fooling yourself

Practice gel answer: Suspect **B** matches the scene. All three of B's bands sit at the same heights as the scene's bands. Suspect C shares one band (the top one) but not the other two, so C is **not** a match. Suspect A matches none.

CHECK YOUR DRAWING ON P. 6A

1. The smallest pieces are drawn farthest from the wells, near the + end.
2. Every lane has the same number of bands as on the iPad, at the same heights.
3. Your match lines up for **every** band, not just one.
4. Your reasoning says why: the same cuts mean the same DNA spelling in those places, and pieces of the same size run the same distance.

A match shows whose DNA was there. It does not prove who did it. The owner of the shed, or a friend who visited, could leave DNA for an innocent reason. Some cases on the iPad end with no match at all; that rules every suspect out.

Why PCR comes first: a drop of blood holds too little DNA to see on a gel. PCR copies it, then the enzyme cuts the copies.

WORDS TO KNOW HERE

restriction enzyme · gel electrophoresis · band · PCR

DONE EARLY?

Tap *Try another case*. Find a case where nobody matches, or one where the match had a good reason to be there.

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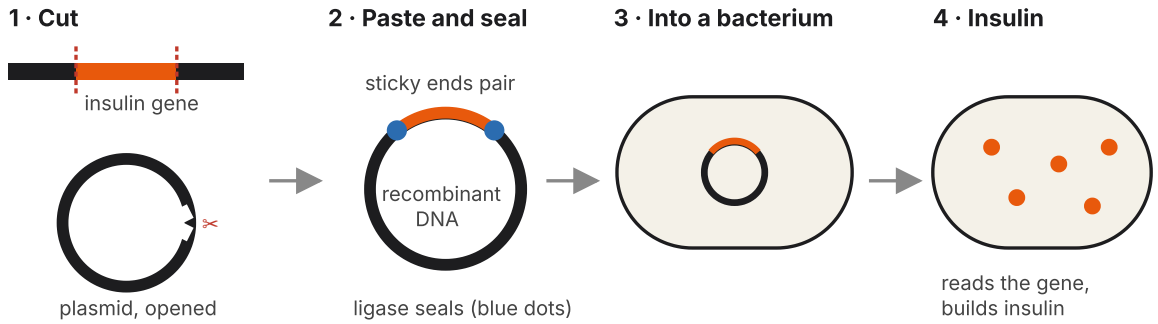
Cut, paste, seal: insulin from bacteria

Job: put a human gene into bacteria so they make a human protein.

QR Toolbox #plasmid (optional)

Treat someone: **3 · Paste a gene** | 4 · Edit a gene

One restriction enzyme cuts both pieces, so the cut ends match.



People with type 1 diabetes need insulin every day. Since 1982, most of that insulin has come from bacteria that carry the human insulin gene.

A **restriction enzyme** cuts the insulin gene out of human DNA and opens a **plasmid**, a small ring of bacterial DNA. Because one enzyme made both cuts, the ends are **sticky ends** that fit each other, like puzzle tabs. **Ligase** seals the joints. The ring now holds DNA from two sources: **recombinant DNA**. A bacterium takes in the ring, reads the gene, and builds insulin.

That is **genetic engineering**, and the bacterium is now a **GMO**, a genetically modified organism.

BUILD IT WITH POP BEADS

1. The white ring (20 beads) is the plasmid. The red chain (10 beads) is the insulin gene.
2. Pop the white ring open at any one joint. That is the enzyme's cut.
3. Snap the red chain into the gap at both ends. Each snap is ligase sealing a sticky end. Count the beads in the new ring.
4. Reset before you leave: pop the red chain out and close the white ring.

WRITE IT DOWN · PACKET P. 6A, BOX 3

Number the four steps in order. On the small ring drawing, label a sticky end and the place where ligase seals.

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Count the beads

Your new ring: 20 white + 10 red = **30 beads**, in one closed ring. That ring is recombinant DNA: a bacterial plasmid holding a human gene.

THE FOUR STEPS, IN ORDER (P. 6A, BOX 3)

1. A restriction enzyme cuts the insulin gene out and opens the plasmid.
2. Ligase seals the gene into the plasmid.
3. A bacterium takes in the recombinant plasmid.
4. The bacterium reads the gene and makes insulin.

Why one enzyme for both cuts? The enzyme leaves the same short, single-stranded tail on every end it cuts. Ends cut by the same enzyme pair up by matching bases. Ends cut by two different enzymes would not fit.

Where the bacteria come in: one bacterium with the plasmid divides again and again, and every copy makes insulin. A big tank of them makes insulin for thousands of people.

WORDS TO KNOW HERE

restriction enzyme · plasmid · sticky ends · ligase · recombinant DNA · genetic engineering · GMO

DONE EARLY?

A scientist wants to check that the plasmid really got the gene. The new plasmid is bigger. Which station's tool would show that, and would the bigger plasmid run farther or less far?

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Scissors with a GPS

Job: find one exact spot in a person's DNA and cut it there.

QR
Toolbox
CRISPR
#crispr

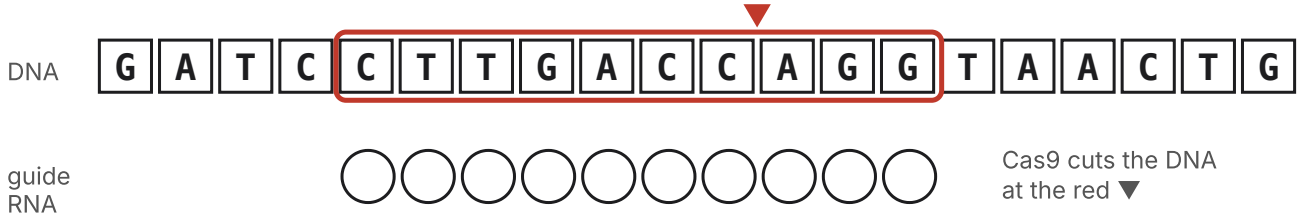
Treat someone: 3 · Paste a gene | **4 · Edit a gene**

Sickle cell disease bends red blood cells into stiff curves that jam small blood vessels and cause pain. Before birth, a baby's red cells carry a different kind of hemoglobin, **fetal hemoglobin**, and those cells stay round. After birth, a **switch gene** turns fetal hemoglobin off.

CRISPR started in bacteria as a way to remember viruses. Scientists turned it into a gene editor with two parts. The **guide RNA** is a short RNA whose bases pair with one spot in the DNA: A with U, T with A, C with G. **Cas9** is the scissors. It cuts only where the guide RNA pairs.

The guide RNA gives the address; Cas9 cuts at that address and nowhere else.

A stretch of the switch gene. The guide RNA must pair with the boxed bases. RNA has U where DNA has T.



MATCH THE GUIDE · ON THE IPAD

1. Scan the QR (or use today's Start Here link). Slide the guide RNA along the DNA until every base pairs. Cas9 cuts there.
2. Watch what the cut does to the switch, and what the red cells do next.

KEEP A RECORD · PACKET P. 6A, BOX 4

Write the guide RNA under the DNA strip, one base per circle. Draw a line where Cas9 cuts. Then finish: after the cut, the patient's cells turn ... back on.

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Check your guide RNA

DNA (boxed)	C	T	T	G	A	C	C	A	G	G
guide RNA	G	A	A	C	U	G	G	U	C	C

The red line is the cut, between the 7th and 8th boxed bases.

Common slip: writing T in the guide. RNA has no T. Where the DNA has A, the guide RNA has **U**.

WHAT THE CUT DOES

The cut breaks the switch gene, so the switch can no longer turn fetal hemoglobin off. The patient's cells make **fetal hemoglobin** again, and the red blood cells stay round. The first approved CRISPR medicine, in 2023, works this way: doctors edit the patient's own blood stem cells outside the body, then put them back.

Why so careful about the match? A guide that paired with the wrong spot would send Cas9 to cut a different gene. Scientists choose a guide that matches only one place in the whole genome.

WORDS TO KNOW HERE

CRISPR · guide RNA · Cas9 · genetic engineering · complementary

DONE EARLY?

Station 3 added a gene. Station 4 cut a gene. Tell your group one way they are alike and one way they are different.

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